Venkata Lakshmi Parimala Pasupuleti

AI Engineer · LLM Applications · Agentic Frameworks · AI Evaluation · Financial Services AI · Rapid Prototyping · Full-Stack Python · Production

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AgentProbe (live eval toolkit) | Fairfax, Virginia · Open to Hybrid NYC/NJ

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production experience building and shipping **LLM applications, agentic frameworks, and RAG pipelines** across financial services and regulated industries -- **PoC to production** at startup speed. Currently at Intuit integrating Anthropic Claude into **financial services AI** products: own the full loop from use case discovery through rapid prototyping, structured evaluation, and production deployment with guardrails and CI eval gates. Built **AgentProbe**: an open-source **AI evaluation / bake-off toolkit** routing across 5 LLM providers (Anthropic, OpenAI, Azure, Groq, Ollama) with a 5-metric LLM-as-judge engine -- used it to run structured head-to-head provider comparisons and cut ~40% LLM inference cost by routing per task. Built **Leasel**: 8-agent LangGraph platform shipped to production with agentic frameworks (LangGraph), RAG, and HITL -- 93.6% precision across 1,240 test cases. 7+ years at Entrata shipping 20+ modules across 120+ sprints with Legal, Compliance, and Operations in a regulated environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

LLM Applications & AI Evaluation | Agentic Frameworks & LangGraph | Financial Services AI | PoC to Production | Rapid Prototyping | RAG & Vector Databases | Guardrails & Responsible AI | Stakeholder Communication | Python & FastAPI | Docker & Kubernetes | AWS & Cloud-Native AI | Full-Stack (Python + React)

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Financial Services AI · LLM Applications · FastAPI · Python · AWS

- Build and ship **financial services AI applications** end-to-end using Anthropic Claude: natural-language query answering, financial data summaries, and reporting features -- **prototype at speed**, evaluate with LLM-as-judge, then shepherd through production with guardrails, CI regression gates, and OpenTelemetry monitoring. Present impact to engineering and product leadership.
- Run **AI evaluation / bake-offs**: structured LLM-as-judge scoring with golden datasets to select between models and detect quality regressions; CI-integrated gates block production deploys when quality drops -- caught 3 prompt regressions before release, zero hallucination incidents post-deploy.
- Integrate **RAG pipelines** grounding Claude responses in customer financial data (similarity search + retrieval); add guardrails (input validation, sensitive-data redaction, output schema enforcement, fallback paths); build ML pipelines (scikit-learn, XGBoost) and React/TypeScript front ends; deploy to AWS via GitHub Actions CI/CD.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · React · AWS · PostgreSQL · Regulated Domain

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints in a regulated financial domain; **communicated impact to senior business and technology stakeholders** across Legal, Compliance, Operations, and Product; reviewed 300+ PRs; mentored junior engineers.
- Built Python/Bash **automation tools** eliminating 80% of manual operational work; built internal debugging tools cutting triage time 40%; optimized 200+ SQL queries (30% gain) via execution plan analysis and indexing.
- Owned HUD income-limit ETL pipeline for 7 years: few thousand federal rows → 180K domain records with validation, rollback checks, compliance enforcement -- zero violations across 7 annual cycles; AWS S3 PHI migration (Libsodium encryption, 100% compliance, zero incidents); Docker containerization cutting execution time 50%.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · PyTorch · HuggingFace · Statistical Modeling

- Built predictive models achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection; fine-tuned classification model (HuggingFace + PyTorch, presented to faculty panel). Mentored 600+ students across Python and database design.

PORTFOLIO PROJECTS

AgentProbe — AI Evaluation Toolkit & Multi-Provider Bake-Off Engine

AI Evaluation · LLM-as-Judge · Agentic Frameworks

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built a structured **AI evaluation / bake-off engine**: LLM Router dynamically routes across 5 providers based on task classification and cost-performance policy -- used at Intuit to run head-to-head provider comparisons for financial query types; ~40% LLM inference cost reduction vs always routing to highest-capability model.
- **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance); CI-integrated regression detection blocks production deploys on quality drops; 3 LangGraph **agentic framework** agents (Orchestrator, RAG Tracer, Auto Fix) for multi-step evaluation. Live at github.com/venkatalakshmiparimala/AgentProbe.

Leasel — Production Agentic AI with RAG & Regulated-Domain Delivery

PoC to Production · Agentic Frameworks · RAG

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Prototyped and **delivered to production**: 8-agent LangGraph platform for compliance teams across 3 surfaces (Document Intelligence, Support Investigation, Tenant Self-Service); responsible AI enforced in code -- LLM reasons, human or deterministic logic decides every regulated call; safety boundary adversarially tested.
- **RAG pipeline** (TF-IDF + cosine reranking + section-aware chunking): 100% citation correctness, 0% unsupported-claim rate; AgentProbe eval harness: 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR — Clinical NLP Prototype to Production (HIPAA)

Rapid Prototype · Regulated Domain

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- Rapid prototype to production: NLP pipeline (LLMs + BERT + embeddings) translating English/Spanish clinical questions into FHIR API calls; HIPAA-compliant guardrails (RBAC, PHI output filtering); ~60% reduction in manual lookup time; Docker containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & LLMs: LLM Applications, AI Evaluation, Agentic Frameworks, LangGraph, LangChain, RAG, Vector Databases, Prompt Engineering, Guardrails, Responsible AI, HITL, LLM-as-Judge, PoC to Production, Rapid Prototyping, Emerging AI Technology, Financial Services AI, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Structured Outputs

Retrieval & NLP: RAG Pipelines, Hybrid Search, Similarity Search, ChromaDB, Pinecone, Weaviate, Elasticsearch, TF-IDF, Embeddings, Semantic Search, BERT, HuggingFace, spaCy, Document Intelligence

Python & Full-Stack: Python (9+ yrs), FastAPI, Flask, Django, React, TypeScript, Node.js, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, Java, Spring Boot, PHP, SQL, PL/SQL, Bash

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, Stored Procedures

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, RabbitMQ, Kafka, Celery